

BRD -Customer Supply Chain Performance Analytics

1. Objective

Develop an interactive Power BI dashboard that provides end-to-end visibility into supply chain operations by analyzing sales performance, customer behavior, product profitability, shipping efficiency, and regional performance. The dashboard will enable business users to monitor KPIs, identify operational bottlenecks, and make data-driven decisions.

2. Business Problem

The organization generates large volumes of transactional supply chain data, making it difficult to extract meaningful insights through raw tables alone. Stakeholders lack a centralized reporting solution to monitor sales trends, shipping performance, customer segmentation, and product profitability. This limits their ability to identify performance issues, optimize logistics, and improve overall business efficiency.

3. Scope

Dashboard with 4 key analysis pages:

- Executive overview
- Customer analysis
- Product and sales performance
- Shipping and delivery analysis
- Interactive filtering and drill-down capabilities

4. Data Source

Size : 95.91 MB
Source : kaggle.com
Type : CSV

5. KPIs

- Sales
- Profit
- Orders
- Customers
- Quantity
- Profit Margin(%)
- Late Delivery(%)
- On-time Delivery(%)
- Average Shipping Days
- Cancelled Shipments
- Average Order Value
- Average Revenue per Customer

5. Timeline

Phase	DAYS
Dataset Understanding & BRD	DAY 1
Data Cleaning (Power Query)	DAY 2
Star Schema Modeling	DAY 3
DAX Measures & KPIs	DAY4

Dashboard Design & Development DAY 5

Testing & Validation DAY 6

Documentation & Final Presentation DAY 7

6. Assumptions

- The dataset is accurate and representative of business operations.
- Customer IDs, Product IDs, and Order IDs are unique identifiers.
- Order Date and Shipping Date are valid and complete.
- Sales and profit values are recorded correctly.
- The dashboard will use imported data (Import Mode) rather than DirectQuery.
- Historical data is sufficient for trend analysis.

7. Dataset Overview Summary

The project uses the **DataCo Global Supply Chain Dataset**, which contains transactional data related to customer orders, products, shipping, and sales operations.

Dataset Highlights

- Approximately **180,000+ transaction records**
- **53 business attributes**
- Multiple business domains:
 - Customers
 - Orders
 - Products
 - Categories
 - Departments
 - Shipping
 - Markets
 - Geographic location